

**AMENDMENT NO. 5 JULY 1987**  
**TO**  
**IS : 731-1971 SPECIFICATION FOR PORCELAIN**  
**INSULATORS FOR OVERHEAD POWER LINES**  
**WITH A NOMINAL VOLTAGE**  
**GREATER THAN 1 000 V**  
**( Second Revision )**

[ This amendment takes into account the sequence of carrying out acceptance tests and introduction of double plan of sampling procedure. ]

[ Page 14, Clause **10.1.2.2** (see also Amendment No. 3)] — Substitute the following for the existing clause:

**'10.1.2.2** The insulators selected in accordance with **10.1.2.1** shall be divided approximately into three parts and subjected to the applicable acceptance tests in the following order:

| <i>Part(s) of<br/>Samples</i> | <i>Tests on String Insulator<br/>Units</i>                                 |                                                                                                                        | <i>Tests on Rigid<br/>Insulator</i>       |                                            |
|-------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|
|                               | Type A                                                                     | Type B                                                                                                                 | Type A                                    | Type B                                     |
| First and<br>Second Parts     | <b>10.5, 10.6,<br/>10.9 ( when<br/>specified ),<br/>10.8 and<br/>10.11</b> | <b>10.5, 10.6,<br/>10.9 ( when<br/>specified ),<br/>10.7 or 10.8<br/>( whichever<br/>is applicable )<br/>and 10.11</b> | <b>10.5, 10.6,<br/>10.8 and<br/>10.11</b> | <b>10.5, 10.6,<br/>10.8 and<br/>10.11</b>  |
| Third Part                    | <b>10.5, 10.6<br/>and 10.12</b>                                            | <b>10.5, 10.6,<br/>10.10 and<br/>10.12</b>                                                                             | <b>10.5, 10.6<br/>and 10.12</b>           | <b>10.5, 10.6,<br/>10.10 and<br/>10.12</b> |

( Page 29, Appendix C ) — Substitute the following for the existing appendix:

**APPENDIX C**  
**( Clauses 10.1.2.1 and 10.1.2.3 )**  
**SAMPLING PROCEDURE FOR INSULATORS FOR**  
**OVERHEAD POWER LINES**

**C-1. SCALE OF SAMPLING**

**C-1.1 Lot** — In a consignment, all the insulators of the same type and design manufactured from the same material in the same factory under similar conditions of production shall be grouped together to constitute a lot.

**TABLE 5 SAMPLE SIZE AND ACCEPTANCE NUMBER**  
( *Clauses C-1.2, C-2.1, C-2.2 and C-2.3* )

| LOT SIZE         | FOR DIMENSIONS AND TEMPERATURE<br>CYCLE TEST |                                  |                             |                              | FOR MECHANICAL<br>ELECTRO-MECHANICAL<br>AND POROSITY<br>TEST |                                | FOR GALVANIZING<br>AND PUNCTURE<br>TEST |                                |                                  |
|------------------|----------------------------------------------|----------------------------------|-----------------------------|------------------------------|--------------------------------------------------------------|--------------------------------|-----------------------------------------|--------------------------------|----------------------------------|
|                  | First<br>Sample<br>Size                      | Second<br>Sample<br>Size<br>(it) | Acceptance<br>Number<br>(a) | First<br>Rejection<br>Number | Second<br>Rejection<br>Number<br>(r <sub>2</sub> )           | First<br>Sample<br>Size<br>(N) | Second<br>Sample<br>Size                | First<br>Sample<br>Size<br>(9) | Second<br>Sample<br>Size<br>(10) |
| (1)              | (2)                                          | (3)                              | (4)                         | (5)                          | (6)                                                          | (7)                            | (8)                                     | (9)                            | (10)                             |
| Up to 1000       | 8                                            | 8                                | 0                           | 2                            | 2                                                            | 5                              | 5                                       | 3                              | 3                                |
| 1 001 to 3 000   | 13                                           | 13                               | 0                           | 2                            | 2                                                            | 8                              | 8                                       | 5                              | 5                                |
| 3 001 to 10 000  | 20                                           | 20                               | 0                           | 2                            | 2                                                            | 13                             | 13                                      | 7                              | 7                                |
| 10 001 and above | 32                                           | 32                               | 1                           | 3                            | 4                                                            | 20                             | 20                                      | 12                             | 12                               |

**C-1.2** The number of insulators to be selected from each lot shall depend upon the size of the lot and shall be in accordance with col 1 and 2 of Table 5 and procedure given in C-2.

**C-1.2.1** These insulators shall be selected from the lot at random. In order to ensure the randomness of selection, procedure given in IS : 4905-1968 'Methods for random sampling' may be followed.

## **C-2. NUMBER OF TESTS AND CRITERIA FOR CONFORMITY**

**C-2.1** All the insulators selected at random according to col 1 and 2 of Table 5 shall be subjected to dimensions and temperature cycle tests. The insulators failing to satisfy either of the requirements shall be termed as defectives. The lot shall be considered as conforming to these requirements if the number of defectives found in the sample is less than or equal to corresponding acceptance number given in col 4 of Table 5. The lot shall be rejected if the number of defectives in the same lot is greater than or equal to the first rejection number ( $r_1$ ) given in col 5. If the number of defectives is between the acceptance number and the first rejection number, a second sample of the same size (see col 3 of Table 5) shall be selected from the lot at random and subjected to these tests. The number of defectives in the first sample and second sample shall be combined. If the combined number of defectives is less than the second rejection number( $r_2$ ) given in col 6 of Table 5, the lot shall be considered as conforming to these requirements. Otherwise the lot shall be rejected without further testing.

**C-2.2** The lot which has been found as conforming to the above requirements shall then be divided into two parts, as shown in col 7 and 9 of Table 5. The number of insulators to be tested for mechanical, electro-mechanical and porosity tests shall be in accordance with col 7 of Table 5. The lot shall be considered as conforming to these requirements if no defective is found in the sample and shall be rejected if there are two or more defectives. If there is one defective, a second sample of same size (see col 8 of Table 5) shall be selected at random and subjected to the tests. The lot shall be considered as conforming to these requirements if no defective is found in the second sample; otherwise the lot shall be rejected without further testing.

**C-2.3** The lot which has been found as conforming to the requirements of C-2.1 shall then be tested for galvanizing test and puncture test. For this purpose, the sample size is given in col 9 of Table 5. The lot shall be considered as conforming to these requirements if no defective is found in the sample and shall be rejected if two or more defectives are found in the sample. If there is one defective, a second sample of same size ( see col 10 of Table 5 ) shall be selected at random and subjected to the tests. The lot shall be considered as conforming to these requirements if no defective is found in the second sample; otherwise the lot shall be rejected without further testing.

**C-2.4** The lot shall be considered as conforming to the requirements of acceptance tests if conditions in C-2.1, C-2.2 and C-2.3 are satisfied.

**(ETDC 3)**